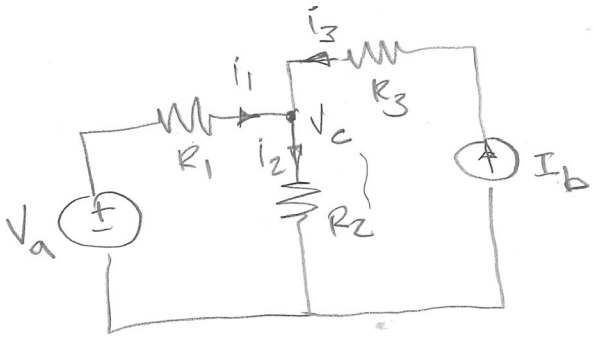
EuoBE

Lösningsförslag

2023-03-16

1.



ANSÄTT NODSPÄNNINGEN V_c I NODEN, ANTAG OCKSÅ STRÖMMAR

FÖR NODEN GÄLLER

$$i_1 + i_3 - i_2 = 0 \quad (1)$$

KSPL GER

$$V_a - R_1 i_1 - R_2 i_2 = 0 \quad (2)$$

(1) & (2)

$$V_a - R_1(i_2 - I_b) - R_2 i_2 = 0$$

$$V_a + R_1 I_b = i_2 (R_1 + R_2)$$

$$i_2 = \frac{V_a + R_1 I_b}{R_1 + R_2} = \frac{5 + 2000 \cdot 2 \cdot 10^{-3}}{2000 + 1000} =$$

$$= 0.003 \text{ A} \quad (3)$$

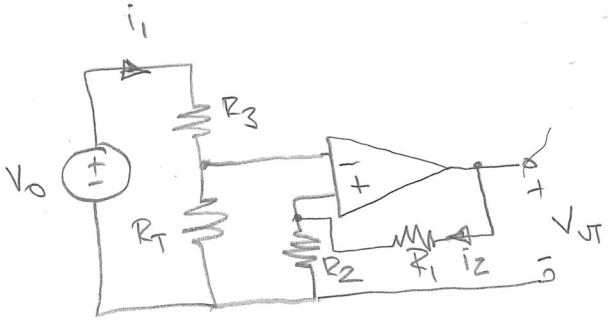
MED (1) FÅS $i_1 = i_2 - I_b = 0.003 - 0.002 = 0.001$ A

OCH EFFEKTEN SOM SP. KÄLLAN AVGER ÄR

$$P_a = V_a i_1 = 5 \cdot 0.001 = 5 \text{ mW}$$

SVAR: $P_a = 5$ mW

2.



$R_T = \alpha(1+\beta T)\ \Omega$

KSPL på R

$$\begin{cases} V_0 - R_3 i_1 - R_T i_1 = 0 & (1) \\ -R_T i_1 + R_2 i_2 = 0 & (2) \\ V_{UT} - R_1 i_2 - R_2 i_2 = 0 \Rightarrow i_2 = \dfrac{V_{UT}}{R_1+R_2} & (3) \end{cases}$$

(3) i (2)

$$i_1 = \frac{R_2}{R_T}\,\frac{V_{UT}}{R_1+R_2} \qquad (4)$$

(4) i (1)

$$V_0 = (R_3+R_T)\,\frac{R_2}{R_T}\,\frac{V_{UT}}{R_1+R_2}$$

ELLER SÅ

$$V_{UT} = V_0\,\frac{R_T(R_1+R_2)}{R_2(R_3+R_T)}$$

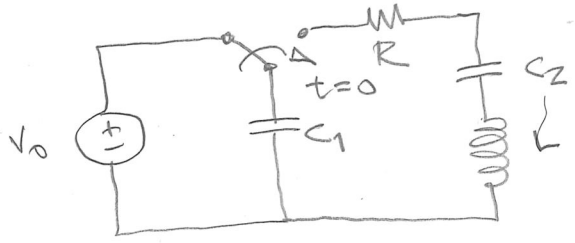
FAST $R_T = \alpha(1+\beta T)$

$$V_{UT} = V_0\,\frac{\alpha(1+\beta T)(R_1+R_2)}{R_2(R_3+\alpha(1+\beta T))}$$

SVAR:

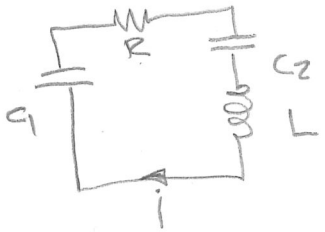
$$V_{UT} = V_0\,\frac{\alpha(1+\beta T)(R_1+R_2)}{R_2(R_3+\alpha(1+\beta T))}$$

3)



Bestäm a) spänning över R för $t \geq 0$

Lösning: för $t \geq 0$ gäller



KSPL ger

$$v_{C_1} + v_{C_2} + Ri + v_L = 0 \quad (1)$$

där

$$\begin{cases} v_{C_1} = v_{C_1}(0) + \frac{1}{C_1}\int i\,dt \\ v_{C_2} = v_{C_2}(0) + \frac{1}{C_2}\int i\,dt \\ v_L = L\frac{di}{dt} \end{cases} \quad (2)$$

(2) i (1) ger

$$v_{C_1}(0) + \frac{1}{C_1}\int i\,dt + v_{C_2}(0) + \frac{1}{C_2}\int i\,dt + Ri + L\frac{di}{dt} = 0 \quad (2.1)$$

DERIVERA MAP t

$$\left(\frac{1}{C_1} + \frac{1}{C_2}\right) i + R\frac{di}{dt} + L\frac{d^2i}{dt^2} = 0$$

$$\frac{d^2i}{dt^2} + \underbrace{\frac{R}{L}}_{2\alpha}\frac{di}{dt} + \underbrace{\frac{1}{L}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)}_{\omega_0^2} i = 0 \quad (2.5)$$

UNDERDÄMPAD LSG ENLIGT RIFORAD GRAF →

LÖSNINGEN BLIR

$$i(t) = e^{-\alpha t}(A\sin(\beta t) + B\cos(\beta t)) \quad (3)$$

DET GÄLLER ATT $i(0) = 0 \Rightarrow B = 0$ (4)

(4) i (3)

$$\begin{cases} i(t) = Ae^{-\alpha t}\sin(\beta t) \\ \beta = \sqrt{\omega_0^2 - \alpha^2} \end{cases} \quad (5)$$

och spänningen över resistansen

$$\begin{cases} V_R(t) = Ri(t) = RAe^{-\alpha t}\sin(\beta t) \\ \beta = \sqrt{\omega_0^2 - \alpha^2} \end{cases} \qquad (5)$$

Vr grafen fås

$$\begin{cases} V_{RTOPP1} = 45\ mm \\ V_{RTOPP2} = 24\ mm \end{cases}$$

(5) ger $\frac{V_{RTOPP1}}{V_{RTOPP2}} = \frac{e^{-\alpha t_1}}{e^{-\alpha(t_1+T)}} = e^{-\alpha t_1 + \alpha t_1 + \alpha T}$

$$\frac{V_{RTOPP1}}{V_{RTOPP2}} = e^{+\alpha T}$$

$$\alpha = \frac{1}{T} LN\left(\frac{V_{RTOPP1}}{V_{RTOPP2}}\right) = \frac{1}{0.31\cdot 10^{-3}} LN\left(\frac{45}{24}\right) \simeq$$

$$\simeq 2027.77$$

men $2\alpha = \frac{R}{L} \Rightarrow L = \frac{R}{2\alpha} \simeq \frac{4}{2\cdot 2027.77} \simeq 1\ mH$

dessutom gäller att

$$\beta = \frac{2\pi}{T} \Rightarrow \omega_0^2 - \alpha^2 = \frac{4\pi^2}{T^2}$$

$$\omega_0^2 = \alpha^2 + \frac{4\pi^2}{T^2} \qquad (6)$$

och ur (2.5) fås, $C_1 = C_2 = C$,

$$\omega_0^2 = \frac{2}{LC} \qquad (7)$$

(7) & (6)

$$\frac{2}{LC} = \alpha^2 + \frac{4\pi^2}{T^2} \Rightarrow C = \frac{2}{L\left(\alpha^2 + \frac{4\pi^2}{T^2}\right)} =$$

$$= \frac{2}{1\cdot 10^{-3}\left(2027.77^2 + \frac{4\pi^2}{(0.31\cdot 10^{-3})^2}\right)} \simeq 4.82\cdot 10^{-6}\ F$$

SVAR:

a) $\begin{cases} U_R(t) = AR e^{-\alpha t} \sin(\beta t) \\ \beta = (\omega_0^2 - \alpha^2)^{1/2} \end{cases}$

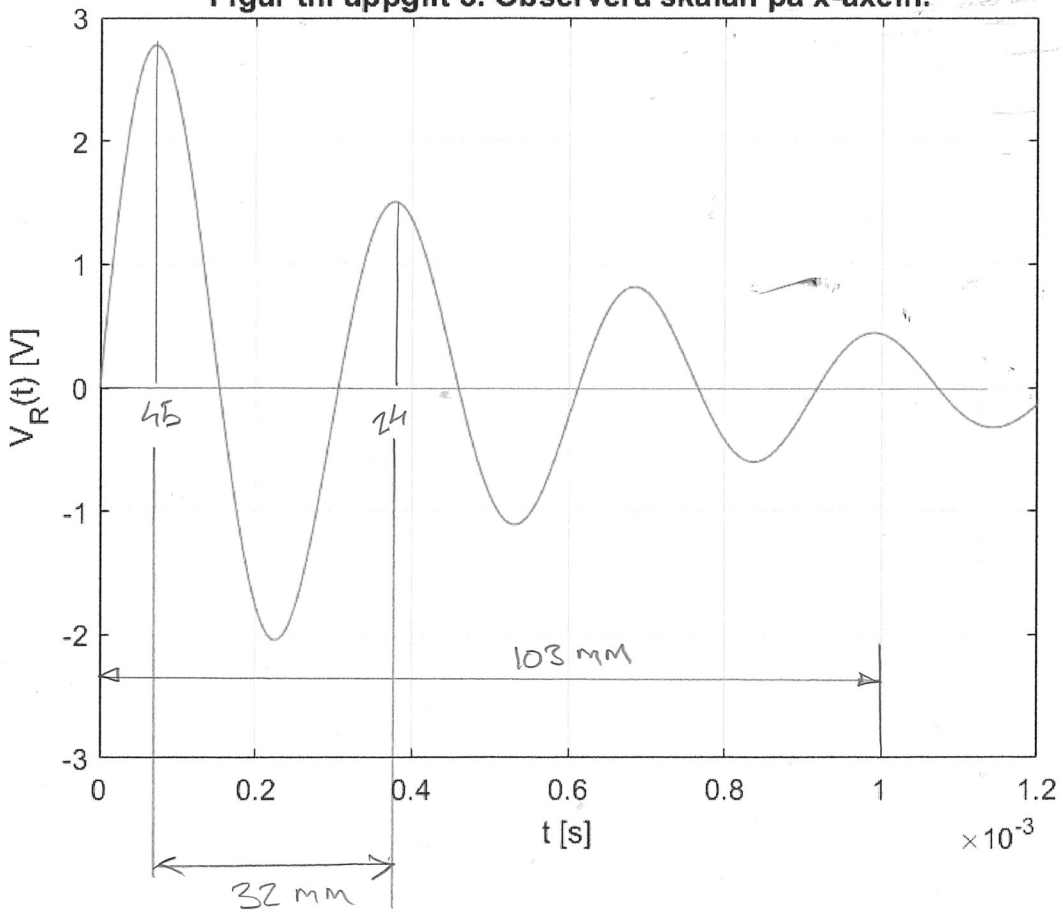
b) $L \simeq 1\ mH$

$C \simeq 4.8\ nF$

FOTNOT: KONSTANTEN A BESTÄMS UR (2.1) VID $t = 0$

NOTERA ATT $i = 0$ VID $t = 0$ DÅ SPOLEN ÄR "STRÖMTRÖG" $i(0^-) = i(0^+) = 0$

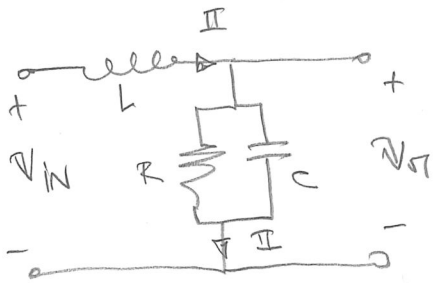
Figur till uppgift 3. Observera skalan på x-axeln.



$1\ \text{ms} = 103\ \text{mm}$

PERIODTID $T = \frac{32}{103} 1 \cdot 10^{-3} = 0.31 \cdot 10^{-3}$ sek

4)



GÅSAR PÅ MED KSPL

$$\begin{cases} V_{IN} - j\omega L\,II - Z_p\,II = 0 & (1) \\ V_{UT} - Z_p\,II = 0 & (2) \end{cases}$$

MEN $$Z_p = \frac{R\,\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{R}{1+j\omega CR} \qquad (3)$$

(3) & (2) $$II = \frac{V_{UT}}{Z_p} = \frac{1+j\omega CR}{R} V_{UT} \qquad (4)$$

(4) i (1) ger

$$V_{IN} - \left(j\omega L + \frac{R}{1+j\omega CR}\right)\frac{1+j\omega CR}{R} V_{UT} = 0$$

$$\frac{V_{UT}}{V_{IN}} = \left[1 + \frac{j\omega L}{R}(1+j\omega CR)\right]^{-1} =$$

$$= \left[\frac{R + j\omega L(1+j\omega CR)}{R}\right]^{-1} =$$

$$= \frac{R}{j\omega L + R(1-\omega^2 LC)}$$

JÄMFÖR SÅ FÅR VI

$$\begin{cases} a = R & c = R(1-\omega^2 LC) \\ b = 0 & d = \omega L \end{cases}$$

OM REHPER NOG, VAD HÄNDER DÅ?

SKRIV OM

$$\frac{V_{UT}}{V_{IN}} = \frac{R\left[R(1-\omega^2 LC) - j\omega L\right]}{(\omega L)^2 + R^2(1-\omega^2 LC)^2}$$

$$\mathrm{Re}\left\{\frac{V_{UT}}{V_{IN}}\right\} = \frac{R^2(1-\omega^2 LC)}{(\omega L)^2 + R^2(1-\omega^2 LC)^2}$$

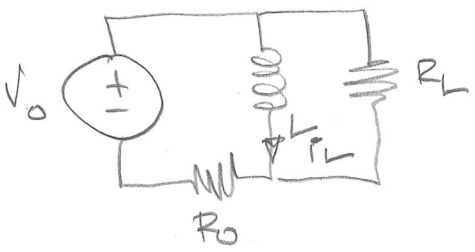
BLIR 0 NÄR OM $(1-\omega^2 LC) = 0$

DVS $\omega_0 = \sqrt{\frac{1}{LC}}$

SVAR: a) $H = \dfrac{R}{R(1-\omega^2 LC) + j\omega L}$

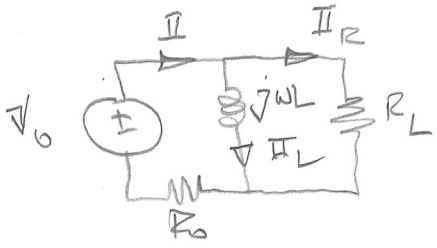
b) $\omega_0 = \sqrt{\dfrac{1}{LC}}$

5.



$i_L = 1$ mA
$f = 20$ kHz
$R_0 = 20$ kΩ
$R_L = 20$ kΩ
$L = 120$ mH

HOPPAR ÖVER TILL KOMPLEXA TALPLANET.
STRÖMMEN $i_L \rightarrow \mathbb{I}_L$ (RIKTFAS)



KSTL I NOD 2

$$\mathbb{I} = \mathbb{I}_L + \mathbb{I}_R \qquad (1)$$

SAMMA SPÄNNING ÖVER L OCH R_L

$$R_L \mathbb{I}_R = j\omega L \mathbb{I}_L \qquad (2)$$

(2) I (1) GER $\quad \mathbb{I} = \mathbb{I}_L + \frac{j\omega L}{R_L}\mathbb{I}_L \qquad (3)$

KSPL I YTTRE

$$\mathbb{V}_0 - j\omega L \mathbb{I}_L - R_0 \mathbb{I} = 0 \qquad (4)$$

(4) & (3) $\quad \mathbb{V}_0 = j\omega L \mathbb{I}_L + R_0\left(1 + \frac{j\omega L}{R_L}\right)\mathbb{I}_L$

RÄKNA LITE SIFFROR

$$\mathbb{V}_0 = \left[R_0 + j\omega L\left(1 + \frac{R_0}{R_L}\right)\right]\mathbb{I}_L =$$

$$= \left[20\cdot 10^3 + j\,2\pi\, 20\cdot 10^3\left(1 + \frac{20\cdot 10^3}{20\cdot 10^3}\right) 120\cdot 10^{-3}\right] 1\cdot 10^{-3}\angle 0^\circ$$

$$\simeq 20 + j\,30.159 \text{ VOLT}$$

$$V_0 = |\mathbb{V}_0| = \left[20^2 + 30.159^2\right]^{1/2} \simeq 36.19 \text{ VOLT}$$

SVAR: $V_0 \simeq 36.2$ ✓

$(\mathbb{V}_0 \simeq 36.2\angle 56.5^\circ \text{ V})$